Unit: Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________

Topic: Linear Inequalities with Two Variables **Date:** ____________

Advanced (Habanero)

Q1. What is the graph of the inequality $x + 2y > 4$?
(A) Solid line with shading above
(B) Dashed line with shading below
(C) Solid line with shading below
(D) Dashed line with shading above
(E) No shading

Q2. Which inequality has a solid boundary line?
(A) $x + y < 2$
(B) $x + y \leq 2$
(C) $x + y > 2$
(D) $x + y \geq 2$
(E) $x - y < 2$

Q3. What is the solution region for the inequality $y < -x + 3$?
(A) Above the line $y = -x + 3$
(B) Below the line $y = -x + 3$
(C) On the line $y = -x + 3$
(D) To the left of the line $y = -x + 3$
(E) To the right of the line $y = -x + 3$

Q4. Which of the following is a graph of the inequality $2x-3y > 5$?
(A) Dashed line with shading above
(B) Solid line with shading below
(C) Dashed line with shading below
(D) Solid line with shading above
(E) No shading

Q5. What type of boundary line is used for the inequality $y \geq 2x - 1$?
(A) Dashed
(B) Solid
(C) Dotted
(D) None
(E) Both

Q6. Which inequality has a dashed boundary line?
(A) $x + y < 2$
(B) $x + y \leq 2$
(C) $x + y > 2$
(D) $x + y \geq 2$
(E) $x - y < 2$

Q7. What is the solution region for the inequality $y > x - 2$?
(A) Above the line $y = x - 2$
(B) Below the line $y = x - 2$
(C) On the line $y = x - 2$
(D) To the left of the line $y = x - 2$
(E) To the right of the line $y = x - 2$

Q8. Which of the following is a graph of the inequality $y \leq -2x + 1$?
(A) Dashed line with shading above
(B) Solid line with shading below
(C) Dashed line with shading below
(D) Solid line with shading above
(E) No shading

Open Ended Questions (FRQ)

Q9. Graph the inequality $x + 2y > 3$ on the coordinate plane and identify the solution region.

Q10. Solve the inequality $y < -x + 4$ and graph the solution region. Identify the boundary line as solid or dashed.

Chef's Tips

- Tip 1: Emphasize the importance of shading and boundary lines.
- Tip 2: Use graph paper to help students visualize the solution regions.
- Tip 3: Encourage students to check their work by testing points in each region.